PostAir Weather API testing (PAT-2)

[PAT-5] BUG-03 | Turbulence report: windSpeed.mph is inconsistent with windSpeed.knots

Created: 29/Oct/26 Updated: 29/Oct/26

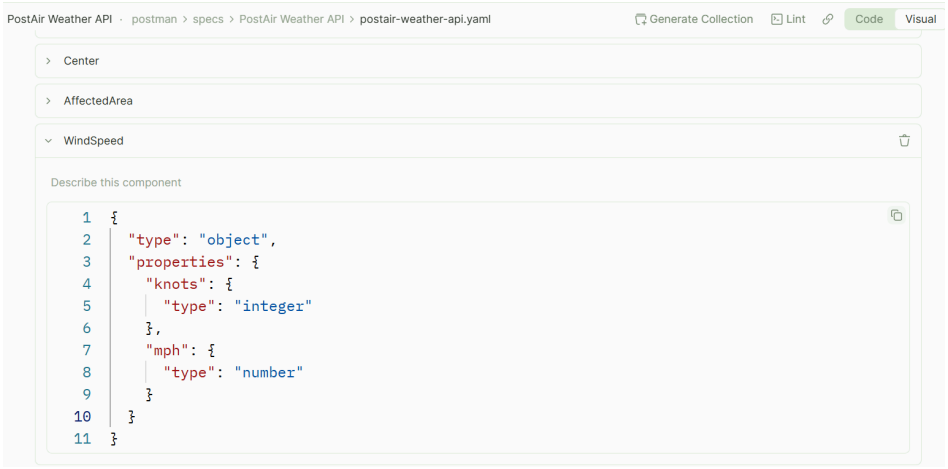
Status:	To Do
Project:	PostAir API Testing
Components:	None
Affects versions:	None
Fix versions:	None
Parent:	PostAir Weather API testing

Type:	Bug	**Priority:**	High
Reporter:	Hagar Samy	**Assignee:**	Unassigned
Resolution:	Unresolved	**Votes:**	0
Labels:	None		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	windSpeed - spec-20261029-023801.png windSpeed - response-20261029-024017.png windSpeed - assertion-20261029-030256.png
Rank:	0|i00049:

Description

Steps	1. Send a GET request with a valid API key to the Turbulence URL in Test Data 2. For each item, compare `windSpeed.knots` with {{windSpeed.mph }}
Expected Result	• `windSpeed.mph` equals `windSpeed.knots` × 1.15078, so `mph` is always larger than `knots` • e.g. 53 kt ≈ 61.0 mph, 43 kt ≈ 49.5 mph, 59 kt ≈ 67.9 mph • A knot is a nautical mile per hour: 1 knot ≈ 1.15 mph

	• **Source**: standard unit conversion: Reference • **Note**: The **spec** (postair-weather-api.yaml, component WindSpeed) defines knots as an integer and mph as a number, with no description and no stated relationship between them.
Actual Result	**98 of 100** items have `mph` that does not match `knots` (tolerance ±1): • `Item 0 (AMD): 53 kt, returned 70.7 mph` • `Item 1 (CCU): 43 kt, returned 53.2 mph` • `Item 2 (MAN): 59 kt, returned 73 mph` Some items have `mph` far **smaller than** `knots`: • `Item 20 (POA): 62 kt, returned 1.6 mph` • `Item 55 (HMO): 84 kt, returned 6.6 mph` ▪ ▪ Same values on repeated runs — The `mph` / `knots` ratio varies between items, so this is not one wrong conversion factor.
Bug Severity	High
Environment	• PostAir Weather API v1 • Postman Collection Runner
Test Data	**Turbulence:** `https://education.postmanlabs.com/v1/weather/turbulence`
Evidence	



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